

Math 204 - Differential Equations (Fall 2025)

Topics to be Covered

Introduction to differential equations, First order ordinary differential equations (ODEs), Second and higher order ODEs, Power series solution of ODEs, The Laplace transform, Systems of linear ODEs, Numerical methods for ODEs, Partial differential equations (PDEs) and Fourier series.

Textbook

Elementary Differential Equations and Boundary Value Problems (12th Edition)
by William E. Boyce, Richard C. DiPrima and Douglas B. Meade

The textbook will be available at the bookstore.

Sections

Section 1, Mon&Wed 8:30 - 9:40 at SNA A43 (Instructor: Sefa Feza Arslan)

Section 2, Mon&Wed 11:30 - 12:40 at SNA B172 (Instructor: Sefa Feza Arslan)

Section 3, Tue&Thr 13:00 - 14:10 at SOS B10 (Instructor: Emre Mengi)

Section 4, Mon&Wed 16:00 - 17:10 at SOS B07 (Instructor: Sefa Feza Arslan)

Instructors

Sefa Feza Arslan (sefaarslan@ku.edu.tr), SCI 231

Office Hours : Mon&Wed 14:30 - 15:30 or by appointment

Emre Mengi (emengi@ku.edu.tr), SCI 267

Office Hours : Tue 15:40 - 17:10 or by appointment

LearnHub

All of the course material will be posted on LearnHub.

You should follow the course mainly through LearnHub. Announcements will be made through LearnHub and/or KUHub.

Grading

Your grade at the end of the semester will be assessed based on two midterms, a final, and quizzes, in particular based on the following grading scheme.

- %30 (Midterm 1) + %30 (Midterm 2) + %40 (Final) + %10 (Quizzes, Bonus)

If your final score is higher than your lowest midterm score, your final score will also replace that midterm score. Please keep in mind that there will be a curve in the end when assessing your letter grade.

The tentative midterm weeks are as follows.

- Midterm 1 : November 10 - November 14 (6th week)
- Midterm 2 : December 15 - December 19 (11th week)

Quizzes

There will be 3-6 quizzes given at the end of the lectures on random dates (you will not know the dates in advance). Each quiz will consist of one or two multiple choice questions related to the topics covered in that particular lecture, and will not take more than 10 minutes. The quizzes are bonus and have %10 weight overall. Every quiz has equal weight.

Problem Sessions

The problem sessions meet once every week. Their purpose is to provide you the opportunity to practice with your teaching assistants (TAs) and classmates, as well as to bring questions/topics that you are having difficulty with. Problem session times are as given below. The TAs teaching the problem sessions will be announced.

PSA, Fri 8:30 - 9:40 at SOS B07

PSB, Fri 16:00 - 17:10 at SOS B07

There will be no problem session during the first week of the semester. The problem sessions will start in the second week of the semester.

Teaching Assistants

Ahmed Choukri Abdullah (aabdullah22@ku.edu.tr; at ENG Z20)

Office Hours: Wed 13:00-14:00

Abdollah Ahmadpour (aahmadpour21@ku.edu.tr; at ENG Z20)

Office Hours: Mon 13:00-14:00

Kaan Hepdurak (khepdurak20@ku.edu.tr; at SCI 130)

Tolga Recep Uçar (tucar24@ku.edu.tr; at SCI 156)

Office Hours: Fri 13:00-14:00

KOLT Tutors

Başak Deniz Altun (baltun22@ku.edu.tr)

KOLT hours : Mon 13:00-14:15, Tue 11:30-12:45, Thr 11:30-12:45

Umut Dalgiç (udalgic23@ku.edu.tr)

KOLT hours : Mon 17:30-18:45, Tue 17:30-18:45, Thr 16:00-17:15

(see also <https://kolttutoring.com/schedule/#MATH204>)

Make-up Exams

A student can be eligible for a make-up exam only if she/he provides proper medical reports approved by the health center at Koç University or an excuse form.

The final exam will be used as the make-up exam for the midterm exams. If a student misses the final exam for a legitimate reason, a separate make-up exam will be held after the final exam period.

Objectives

To provide the student with a basic knowledge of ODEs and PDEs. This includes the general theory of and solution techniques for general first order ODEs, second order linear ODEs, systems of first order ODEs, and the heat equation, if time permits also the wave equation.

Classroom Code of Conduct

Students at Koç University are required to adhere to classroom code of conduct and to refrain

from all forms of unacceptable behavior during lectures. The activities which are prohibited in class include and are not limited to the following:

- Arriving late or leaving early without the prior permission of the professor.
- Engaging in side conversations.
- Using cell-phones or laptops for purposes that are not course-related.

Please read the Koç University Student Code of Conduct from <https://apdd.ku.edu.tr/en/academic-policies/student-code-of-conduct/>.

Course Calendar

This calendar is tentative, and subject to changes. The numbers in parentheses refer to the sections from the textbook by Boyce, DiPrima and Meade.

Week 1 (Oct 6 - 10)

Introduction to Differential Equations, Method of Integrating Factors (1.1-1.3, 2.1)

Week 2 (Oct 13 - 17)

Separable Differential Equations, Autonomous Differential Equations, Existence & Uniqueness of the Solution (2.2, 2.4-2.5)

Week 3 (Oct 20 - 24)

Exact Differential Equations, Second Order Homogeneous Equations with Constant Coefficients (2.6, 3.1)

Week 4 (Oct 27 - 31)

Second Order Linear ODEs, Wronskian (3.2)

Week 5 (Nov 3 - Nov 7)

Complex and Repeated Roots of Characteristic Equation, Second Order Nonhomogeneous Equations, Method of Undetermined Coefficients (3.3-3.5)

Week 6 (Nov 10 - 14)

Variation of Parameters, Review of Power Series (3.6, 5.1)

Week 7 (Nov 17 - 21)

Series Solutions of Second Order Linear Equations, Definition of Laplace Transform (5.3-5.4, 7.1)

Week 8 (Nov 24 - Nov 28)

Solution of Initial Value Problems by Laplace Transform, Step Functions, Discontinuous Forcing Functions, Convolution (7.2-7.4, 7.6)

Week 9 (Dec 1 - 5)

Systems of First-Order Linear Equations: Basic Theory, Homogeneous Systems of ODEs with Constant Coefficients: Distinct Real Eigenvalues (6.1, 6.4-6.5)

Week 10 (Dec 8 - 12)

Homogeneous Systems of ODEs with Constant Coefficients: Complex Eigenvalues, Nonhomogeneous Linear Systems, Numerical Methods for the Initial Value Problem (6.6, 6.9, 8.1-8.2)

Week 11 (Dec 15 - 19)

Runge-Kutta Method (8.3)

Week 12 (Dec 22 - 26)

Two-Point Boundary Value Problems, Fourier Series (10.1-10.3)

Week 13 (Dec 29 - Jan 2)

Cosine and Sine Series, Separation of Variables, Heat Conduction in a Rod (10.4-10.5)

Week 14 (Jan 5 - 9)

IF TIME PERMITS — Other Heat Conduction Problems, Wave Equation (10.6-10.7)

Important Dates and Holidays

- October 6, Mon — First Day of Classes
- October 6 - 17 — Add-Drop Period
- October 29, Wed — Foundation of the Turkish Republic (Oct 28-29, Holiday)
- December 12, Fri — Last Day for Withdrawal from a Course
- January 1, Thu — New Year's Day (Holiday)
- January 9, Fri — Last Day of Classes
- January 12 - 23 — Final Examination Period